BROMIC
HEATING

COMMERCIAL END USER (HORECA)

The outdoor heating planning guide: operator edition

Operator's guide to patio heating: revenue case, five-year fuel cost with the formula, controls by channel, fire-code pointers and multi-site rollout.



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BROMIC HEATING
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The outdoor heating planning guide

Operator edition, for restaurants, hotels, bars and venues. Extend the Moment.

At US average 2026 prices, natural gas is the cheapest fuel per square foot heated and propane the dearest, with electric between them. In a 30-seat, 600 ft² example, five years of propane costs more than twice electric before cylinder handling, because a freestanding heater serves no more than 4 seats; electric costs about three times natural gas. The formula and the worked example are in Step 3. Purchase price is the smallest number in the decision; energy, handling, controls and the fire code decide it.

Outdoor seating is mainstream: in a 2020 National Restaurant Association survey, 74% of full-service operators offered it and 49% were extending the outdoor season, including with tents or patio heaters[1].



Resort loggia with Eclipse pendants at dusk

First: what is a heated seat worth?



Added revenue = heated seats × turns per night × average check × extra operating nights per year × capture rate

Payback (months) = all-in installed cost ÷ (added contribution per year ÷ 12)

Use contribution margin, not revenue, for payback. Count only nights the patio would not sell without heat. Turns and check come from your POS; extra nights from last year's weather closures. The ROI calculator runs it for your site.

The nine steps



STEP	DECISION	TOOL
1	Is it outdoor?	This guide; fire marshal if enclosing
2	Measure seats and zones	Floor plan
3	Choose the fuel on five-year cost	TCO model
4	Choose style and mounting	Range table
5	Size it	How many heaters? calculator
6	Plan power or gas	Electrician, gas fitter
7	Controls: zones and schedules	Affinity Smart-Heat Controls
8	Clearances, fire code and permits	Local AHJ and fire marshal
9	Design service and rollout	Bromic design service (A07)

Step 1. Is it outdoor?



Outdoor means at least 30% of the perimeter permanently open; a partial roof or awning holds a heat cloud that adds to the radiant heat[3]. Walls, glazing or plastic sidewalls can change both the heating design and what the fire code permits (Step 8). Radiant heat warms guests and surfaces, not air, so it works in a light breeze; forced air does not work outdoors[3].

- ☐ Open perimeter 30% or more[3]; enclosure plans noted
- ☐ Covered, partly covered or open

Step 2. Measure seats and zones



- ☐ Heated seats, tables and bar rail drawn; service aisles left out
- ☐ One zone per section that opens and closes on its own; ft² per zone
- ☐ Ceiling, canopy or wall heights; supply voltage (208, 240 or 277 V)[3]; gas on site

Step 3. Choose the fuel on five-year cost

5-year TCO = hardware + controls + installation + permits + 5 × (energy + handling + maintenance)

Energy per hour: electric = kW × $/kWh · natural gas = Btu/h ÷ 100,000 × $/therm · propane = Btu/h ÷ 91,500[23] × $/gal.

Rates used, the same as the A20 TCO model: electricity 13.54¢/kWh (EIA, US commercial average, May 2026)[18]; natural gas $0.87 per therm (EIA, US commercial, January–May 2026 average)[19]; propane $2.674 per gallon (EIA, US residential, week of 30 March 2026; EIA publishes no commercial series)[20].

HEATER AT FULL OUTPUT	COVERS	COST PER HOUR
Tungsten Electric 4000 / 6000 W	130 / 160 ft²	$0.54 / $0.81
Platinum Electric 3400 / 4500 W	108 / 143 ft²	$0.46 / $0.61
Eclipse Electric 2900 W	121 ft²	$0.39
Platinum Gas 39,800 Btu/h, natural gas / propane	215 ft²	$0.35 / $1.16

HEATER AT FULL OUTPUT	COVERS	COST PER HOUR
Tungsten Freestanding Gas 38,000 Btu/h, natural gas / propane	215 ft²	$0.33 / $1.11

Worked example: 30 seats, 600 ft², 900 heater hours a year

30 seats × 20 ft² = 600 ft²[28], partly covered, cool (factor 0.85; Step 5). 900 hours (6 hours × 150 nights) is the A20 model's placeholder, not a benchmark.

OPTION	HEATERS	ENERGY A YEAR	ENERGY OVER 5 YEARS	CONTROLS
Tungsten Electric 6000 W	600 ÷ (160 × 0.85 = 136) = 4.4 → 5	5 × 6 kW × 900 h × $0.1354 = $3,656	$18,279	5-channel Affinity Dimmer, one heater per 6 kW channel (5 channels); wall switch; app
Platinum Gas 39,800 Btu/h, natural gas	600 ÷ (215 × 0.85 = 182.75) = 3.3 → 4	4 × 0.398 therm × 900 h × $0.87 = $1,247	$6,233	One Affinity On/Off channel (up to 10 gas heaters); wall switch; app
Tungsten Freestanding Gas, propane	larger of 3.3 → 4 by area and 30 seats ÷ 4 = 7.5 → 8	8 × 0.4153 gal × 900 h × $2.674 = $7,996	$39,979, plus handling	

Energy only, at full output; your bill, schedules and dimming lower it. Propane adds about 2,990 gallons a year, about 636 swaps of a 4.7-gallon cylinder, to buy, swap and store. Run hardware, installation and handling in the TCO model. The 2021 guide's per-hour figures used 2021 prices and are retired.

	ELECTRIC	NATURAL GAS	PROPANE
Handling	None	None after install	Cylinder logistics and storage
Control	1% dimming, zones, schedules, app, BMS	On/off, zones, schedules, app, BMS	Freestanding: confirm
Fire code	Simplest path	Fixed, listed outdoor install	Most restricted (Step 8)
Gas restrictions	Compliant	Check local rules (A12)	Check local rules (A12)

DIAGRAM: Stacked bar chart, five-year cost per 1,000 ft² heated for electric, natural gas and propane, split into hardware, install, controls, energy and handling, from the A20 model once rates are verified.

- [ ] Fuel per zone chosen on five-year cost
- [ ] Your own tariff entered in the TCO model

Marina-side bar with heaters under a canopy

Step 4. Choose style and mounting

FAMILY	BEST COMMERCIAL USE	WARRANTY
Tungsten Smart-Heat Electric, 2000–6000 W	Canopies, soffits, walls; recessed	2 yr heater, 7 yr elements[4]
Platinum Smart-Heat Electric, 2300–4500 W	Canopies, awnings, walls, poles	3 yr[5]
Platinum Smart-Heat Electric Marine	Waterfront, marina and vessel decks; IPX5, Lloyd's Register approval	5 yr land, 3 yr vessel[6]
Eclipse Smart-Heat Electric	Over tables: heat and dimmable light in one	3 yr[7]
Eclipse Smart-Heat Freestanding	Movable heat and light for flexible layouts	3 yr[8]
Cobalt Smart-Heat Electric, 4000 / 6000 W	Back-of-house and service areas, walls and ceilings	1 yr parts, 7 yr elements[9]
Platinum Smart-Heat Gas, 23,600 / 39,800 Btu/h	Larger open terraces, fixed gas	2 yr excl. hoses, igniters[10]
Tungsten Smart-Heat Gas built-in, two sizes	Larger open terraces, fixed gas; pivot arm	2 yr excl. hoses, igniters[11]
Tungsten Smart-Heat Freestanding Gas	Open terraces; propane or natural-gas quick-connect	1 yr (propane)[12]

FAMILY	BEST COMMERCIAL USE	WARRANTY
Tungsten Electric Coastal (coming soon)	Waterfront venues; 316 stainless	2
Tungsten Electric Select (coming soon)	Design-led venues; Black, White, Graphite	2

- [ ] One heater family per site type (canopy, wall, freestanding)
- [ ] Coming-soon models carry an approved current alternate

Wall-arm Eclipse heaters over a restaurant terrace

Step 5. Size it



Heaters per zone = zone ft² ÷ (coverage × cover factor × climate factor), rounded up. Cover: open 0.85, partly covered 1.0, covered 1.1. Climate: mild (above about 50°F) 1.0, cool (about 40–50°F) 0.85, cold (often below about 40°F) 0.7. Freestanding heaters also serve no more than 4 seats each, because guests block their radiant heat: take the larger of the two counts. Without a seat count, seats = ft² ÷ 20[28].

HEATER	COVERAGE PER HEATER
Tungsten Electric 2000 / 3000 / 4000 / 6000 W	65 / 100 / 130 / 160 ft²[4]
Platinum Electric and Marine 2300 / 3400 / 4500 W	65 / 108 / 143 ft²[56]
Eclipse and Eclipse Freestanding	121 ft²[2]
Cobalt Electric 4000 / 6000 W	130 / 160 ft²[9]
Platinum Gas 23,600 / 39,800 Btu/h	160 / 215 ft²[10]
Tungsten Gas built-in, smaller / larger	160 / 215 ft²[11]
Tungsten Freestanding Gas, 38,000 Btu/h	215 ft²[2]
Tungsten Electric Coastal and Select (coming soon)	64–144 ft²[2]

Coverage is approximate and varies with mounting height, environment and installation[2].

- ☐ Count per zone; calculator run for each site

Step 6. Plan power or gas



HEATER	AMPS AT 208 V	AMPS AT 240 V
Platinum Electric 2300 / 3400 / 4500 W	11.1 / 16.3 / 21.6	9.6 / 14.2 / 18.8[5]
Eclipse Electric 2900 W	13.9	12.1[7]
Cobalt Electric 4000 / 6000 W	18.75 / 28.4	16.6 / 25[9]
Tungsten Electric 4000 / 6000 W (calculated, W ÷ V)	19.2 / 28.8	16.7 / 25.0

Tungsten 4000 and 6000 W are also offered at 277 V[4]. Built-in gas heaters need a gas line and a 110 V, 2 A supply[1011]. Circuits are sized by a licensed electrician to NFPA 70[21]; gas by a licensed fitter to NFPA 54[22].

- ☐ Panel capacity and circuits per zone
- ☐ Gas load and shut-offs; 110 V at each built-in gas heater

Step 7. Controls: zones and schedules (required)

Controls make the Step 3 numbers real. Without them, heat runs at full output until someone switches it off.

NEED	CONTROL	CAPACITY
Heat one section while another is closed	Affinity On/Off or Dimmer controller, one channel per zone	1, 2 or 5 channels; 6 kW per channel at 208–277 V[1314]
Gas heaters on a schedule	Affinity On/Off controller	Up to 10 gas heaters per channel[13]

NEED	CONTROL	CAPACITY
Eclipse over tables	Affinity Eclipse controller	One Eclipse per channel; 1 or 5 channels[15]
Lower output on mild nights	Dimmer (1% steps) or Eclipse steps (33/66/100%)	[147]
Staff override at the station	Manual wall switch	20 A, 120–277 V[16]
On before service, off at close	Bromic Smart-Heat app: zones, schedules, timers	Through a compatible Affinity controller
Venue-wide control	Expansion board: Ethernet and 0–10 V to a BMS (BMS mode disables the app)	Built into the 5-channel[17]

Wi-Fi and app features require compatible Affinity Smart-Heat Controls, sold separately. Control4, Crestron and Lutron compatibility varies and may need extra hardware or configuration[2].

- [ ] Zone-to-channel schedule
- [ ] Schedules tied to service hours; wall switch for the closing manager
- [ ] Wi-Fi at the controller, or BMS integrator named

Eclipse over a rooftop table, city lights beyond

Step 8. Clearances, fire code and permits



Start with the fire marshal. Rules are adopted and amended locally.

TOPIC	PRIMARY SOURCE	POINTER
Mounting heights	Product pages	Electric wall 72 in, ceiling 96 in (Cobalt 6 ft / 8 ft); Eclipse 96 in; built-in gas 96 in[45971011]
Portable unvented fuel-fired heaters	International Fire Code 603.4[26]	Prohibited in Group A (assembly, including restaurants) and other occupancies; outdoor gas-fired heaters fall under their own subsection[24]
Outdoor portable gas heaters	IFC 603.4.2 as locally adopted	One interpretation bars them inside tents, membrane structures and on balconies, and requires 5 ft from buildings, combustible overhangs and exits[25]
Enclosed patios	IFC 603.4 interpretation	North Carolina: a patio enclosed with clear plastic needs a vented fuel-fired heater or an electric heater[24]
LP-gas cylinder storage	NFPA 58[27]	Storage location, quantity, handling
Electric installation	NFPA 70[21]	Circuits, disconnects, wiring methods
Gas installation	NFPA 54[22]	Fixed supply, clearances, ventilation
Certification	Intertek, UL or CSA mark	Required on gas and electrical appliances, including portable heaters[3]

This table points to primary sources; it is not legal advice. The locally adopted code and the AHJ's interpretation govern. Confirm with the local building and fire authority, and use a licensed electrician or gas fitter.

- [ ] Fire-marshal and building-permit path confirmed per site
- [ ] Clearances per installation manual

Step 9. Design service and rollout

The Bromic design service turns photos, plans or drawings into heater selection, placement and coverage drawings, at no charge[2].

1. **Standardise the package:** one heater family, one controller set-up, one mounting detail per site type.
2. **Survey every site the same way:** Steps 1–8 checklists.
3. **Clear permits in parallel;** plan around the slowest jurisdiction.
4. **Pilot, measure, scale:** covers and sales on heated nights against the ROI case.
5. **Name an owner per site** for schedules and seasonal changeover.

- [ ] Layout requested for each site type
- [ ] TCO (A20) and ROI (A21) run per site

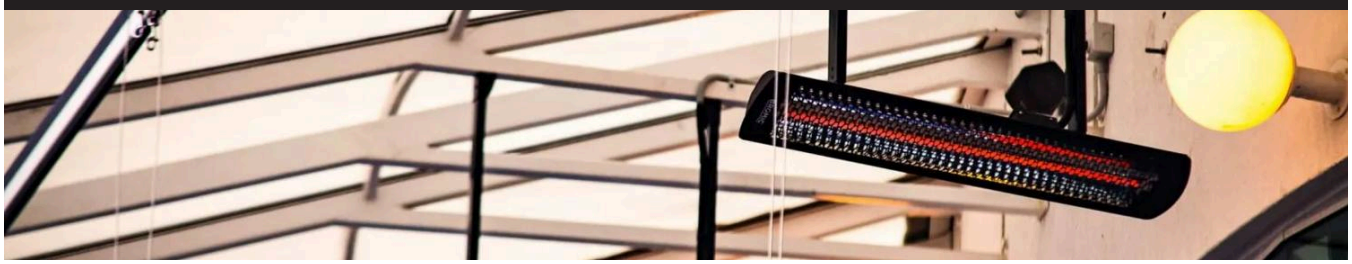
Tungsten freestanding gas beside a set dining table

Guests dining on a heated restaurant terrace

Get your ROI assessment and see what heated seats are worth at your site.

NEXT STEP

Get your ROI assessment



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Confirm every installation with your local building authority and a licensed electrician or gas fitter. Wi-Fi and app features need compatible Affinity Smart-Heat controls, sold separately. © 2026 Bromic Heating · bromic.com